

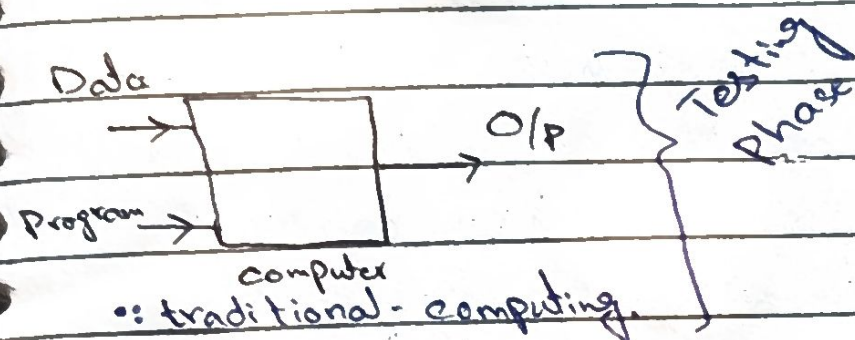
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POST MID

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→ Towards machine learning

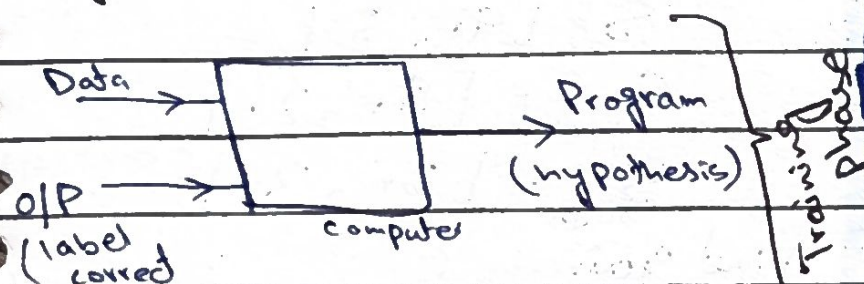
→ It's a bottom-up approach



It is top-down approach and it fails sometimes.

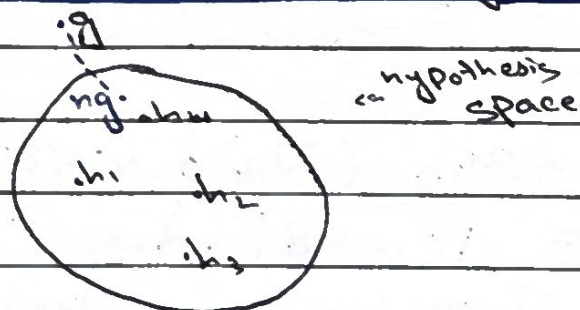
To cover this we use

bottom-up approach to fix the issue and instead of program/heuristic we give it O/P



∴ Machine learning

→ After training it, we give the data, hypothesis to the computer and detect the output in testing phase.



⇒

in ML we

Find nearest to the true thing like h_g is nearest to g

Machine learning

1- Supervised ML

2- Unsupervised ML

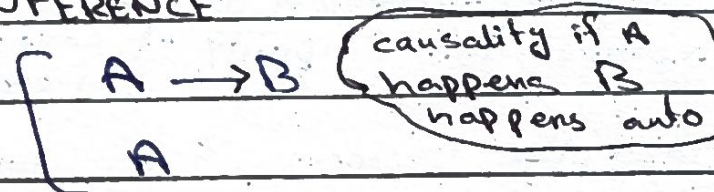
3- Reinforcement learning (RL)

*1 2 types: ^{Discrete} classification, ^{Continuous} Regression

*2 we cluster things data in it, if we find a type & cluster it builds it.

*3 Run a cycle of explore and exploit and rewards the model. e.g., like training a dog and giving biscuit to it.

INFERENCE

⇒ Modus Ponens

INDUCTIVE LEARNING

From repeated experience machine learns and it gives output and all major models are on inductive learning. Machine learning is "inductive" in nature.

'g' is the truth we are finding and it does not exist. and if we are saying our hypothesis is 100% right so our 'g' is above 'h_g'. It is not possible. It searches from the set of hypotheses

we have to calculate the gain of each attribute
so to get to know which attribute is playing
its part more in answer.

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Information gain is the expected reduction
in entropy caused by partitioning the eg.

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

if one subset is all true and
the other subset is all false
the entropy for both is 0.
So after inserting values in
the formula of gain the
term gets '0' which is of
summation. So entropy remains
maximum.

Example:

Values(wind) = weak, strong

$$S = [9+, 5-]$$

$$S_{\text{weak}} = [6+, 2-]$$

$$S_{\text{high}} = [3+, 3-] \quad \left\{ \begin{array}{l} \text{entropy} \\ \text{max} \end{array} \right.$$

Find gain...

$$\text{Gain}(S, \text{wind}) = \text{Entropy}(S) - \sum_{v \in \{\text{weak, strong}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$= \text{Entropy}(S) - \frac{8}{14} \text{Entropy}(S_{\text{weak}}) - \frac{6}{14} \text{Entropy}(S_{\text{high}})$$

$$= 0.940 - \left(\frac{8}{14}\right) 0.911 - \left(\frac{6}{14}\right) 1.00$$

$$= 0.048 \text{ Ans.}$$

→ If we have same value = 1
 $\text{entropy}[1+, 1-] = 0$

→ if we have only one value for to
calculate entropy it's zero.

$$\text{entropy}[1+, 0-] = 0; \text{Entropy}[0+, 1-] = 0$$

"Biased". ML is biased in selection
of hypothesis.

Decision tree:

Disjunction logic

Injunction logic

ID3 - ALGORITHM.

$$\text{Entropy}(S) = -p_+ \log_2 p_+ - p_- \log_2 p_-$$

It is a measure of randomness, uncertainty.

S_+ = classes which are labeled Yes

S_- = classes in which are labeled No

p_+ → measures the impurity of collection of example

→ For the problem 3.2 for

tennis playing

uses entropy to measure and helps to solve problem.

$$|S| = 14; n_+ = 9; n_- = 5$$

$$p_+ = \frac{n_+}{|S|} = \frac{9}{14} = 0.642$$

$$p_- = \frac{n_-}{|S|} = \frac{5}{14} = 0.357$$

$$E(S) = -(0.642) \log_2 (0.642) - \frac{5}{14} \log_2 (0.357)$$

$$= -0.2845 - (-0.3677)$$

$$= 0.2845 + 0.3677$$

$$= 0.08322 \text{ Ans.}$$

Exclude 3.5, 3.6

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ID3 Algorithm.

- create a root node for tree.
- if all e.g. +ve, return single-node root tree +ve.

- if all e.g. -ve, return single-node root tree with label -ve.

- if attribute is empty, return the label of the upper node.

Solve Ques.

id	class	a1	a2
1	+	T	T
2	+	T	T
3	-	T	F
4	+	F	F
5	-	F	T
6	-	F	T

Total entropy(S) = $\left[\frac{3}{6}, \frac{3}{6} \right] = 1.00$

entropy a1:

→ For true = $\left[\frac{2}{3}, \frac{1}{3} \right]$

$= -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.9183$

→ For false: $\left[\frac{1}{3}, \frac{2}{3} \right]$

$= -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = 0.9183$

Gain(S, a1) = Entropy(S) - $\frac{3}{6}$ Entropy(S_T) - $\frac{3}{6}$ Entropy(S_F)

$= 1 - 0.9183 \times \frac{3}{6} - 0.9183 \times \frac{3}{6}$

$= 0.0817$

Now for a2:

3.7 Issues in Decision Tree Learning

3.7.1: Avoiding Overfitting the Data

Split information (S, A) =

$$-\sum_{i=1}^2 \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|}$$

if we want to calculate gain for each 14 days and check which one is good:

Entropy(S, A=day) = Entropy(S) -

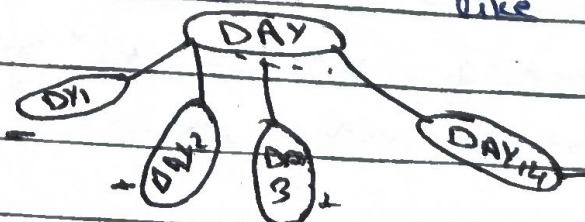
$$\sum_{i=1}^{14} \frac{|S_{v_i}|}{|S|} \text{Entropy}(S_{v_i})$$

⇒ If the entire record is of one class the entropy is '0'

$= 0.94 - \left(\frac{1}{14} (0) + \frac{1}{14} (0) + \dots + \frac{1}{14} (0) \right)$

$= 0.94$ Ans

So the graph would look like this



$$\text{Gain Ratio}(S, A) = \frac{\text{Gain}(S, A)}{\text{Split information}(S, A)}$$

$\Rightarrow$ if $(S_i \approx S)$ for one of the S_i . This either makes the gain ratio undefined or very large for the attributes so alternative to this.

we need to compute arg and find attributes which are above Arg and apply gain ratio on them

Ways to early stop the decision trees

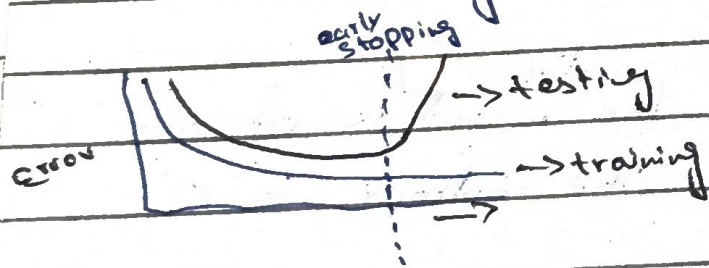
$\rightarrow$ stop at random time and check the error validation and as soon as the graph, validation error goes up stop right there it is called Pre-pruning

$\rightarrow$ It's not always the best solution for decision tree

$\rightarrow$ For ID3 - Post Pruning is prioritize to avoid over-fitting

OVERFITTING DATA

excessive training lead to the overfitting of data



2 methods of Post Pruning:

- 1) Reduce error pruning
- 2) Rule Post Pruning

generalization: Performance of the model on unseen data.

$\rightarrow$ we do early stopping if the testing result is going to explode with errors

the node where we want and make it leaf node and then compare the accuracy

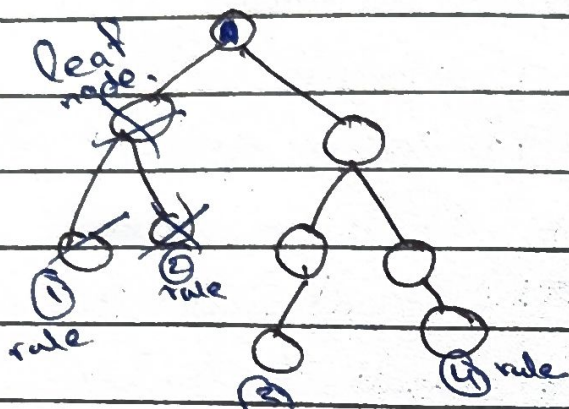
$\rightarrow$ we want information gain to be high.

and output and error and if error reduced its best.

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making overfit tree to the
simple tree and make
it leaf node
and we select the node
whose gain is greater
and we prune it.



- Systematic use bottom-up way for pruning.
- We can also prune on the basis of Gain.

2) Rule Post-Pruning

It is one step ahead of the reduced error ~~pruning~~ pruning.

We write all the rules from the decision tree and change the Pre conditions of the rule and try new combinations and check for accuracy.

$A=A, B=B \rightarrow \text{Yes.}$

Issue For ID-3

→ Instead of continuous values convert it into discrete. and we convert it on the bases of range.

-> Sort the attribute first.

→ But it is not optimal in ID3. I identify those points where the class information is dividing, and make boundary.

Temp →	40	48	60	72	80
Play →	No	No	Yes	Yes	No

The threshold is best which gives us max gain.

→ find entropy

→ Find subgroup entropy.
and Find in which
split we have more
correct info.

→ and in the boundary if we apply testing the majority response, answer will be like that threshold.

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↳ If some value is missing in data set what to do??

Way 1:

Find the most common value in that attribute, Mean, mode or avg.

Way 2:

In the same class if answer is No in weather data set. check that attribute value in those tuples who has same class and fill the empty with that.

→ count occurrence of all other records in other tuples and find probability.
6 = High, 7 = Normal in humidity
total = 14.

$$\frac{6}{14} \quad \frac{7}{14}$$

0.42 0.5
move towards

*

↳ If we have data set of 100 records, make 10 decision trees from it. But one says only 1 can be made from one dataset make 10 from 1 use bootstrapping.

RANDOM FOREST ALGORITHM

These are the model which are used where data set is small and we cannot train deep models.

→ It is a general of decision trees.

→ Its rational is why consult a doctor. Consult 5 doctors for better results.

Ensemble Learning

Bagging Boosting
stacking

Random Forest is an example of Bagging which consist of weak decision tree.

↳ Bootstrapping:-

Pick random samples with replacement from one dataset and make a new one.

→ Repetitions can be around 38%

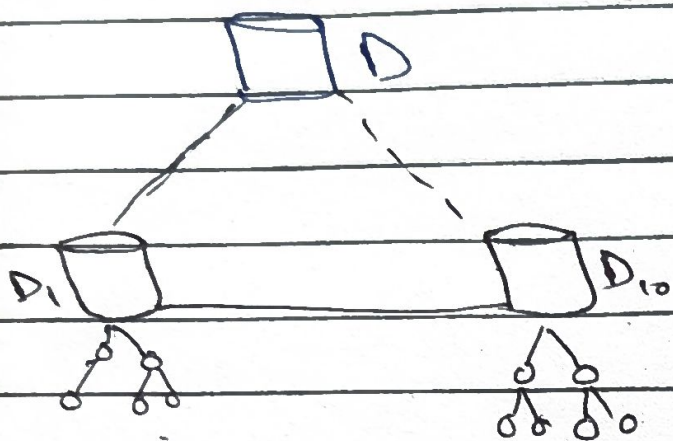
and

60% repetition
40% non-repetition

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and build a decision tree from the sample data and all those trees are called weak decision trees.



and if A_1 is selected at level 1 and we find it on another level. This what random Forest gives us and we cannot do this in ID3.

-> And then select each random features which are participating in decision

-> Why use this than ID-3?

Aw -> We are increasing Data set and training is getting better.

-> In this we end up making similar trees as 60%. is repetition so correlation.

determine we subset size by taking either $\sqrt{F}$, $F/3$ or $\log F$

↳ Sampling is done without replacement.

-> At one level we cant select data with replacement.

But if we go down we select Fresh sample.

/// In the start

Date: // Here.

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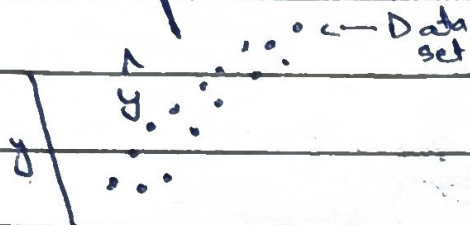
Step size:

→ In linear regression our hypothesis is the vector not the trees.

→ Basically it's the eq of line.

Problem of linear

regression:



$$= \theta_0 x_0 + \theta_1 x_1 + \dots + \theta_d x_d$$

$$= [\theta_0 \dots \theta_d] \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_d \end{bmatrix}$$

it is the linear combination.

It provides a line which in sense considered the best fit line.

what is the best value of θ that the errors residues are minimized. that will be called best fit line.

→ If we make squares of distances it minimize them and take sum of it, it gives the area which show total error {least square line}

→ we start from any random line and then find the solution.

Features.

→ where no of eq is more and variables less it is over determined.

	x_0	x_1	x_2	...	x_d	y output
(x^1, y^1)	1					
(x^2, y^2)	1					
(x^3, y^3)	1					

↑
Samples

$$x_{n \times d+1} \theta_{d+1 \times 1} = y_{n \times 1}$$

$$x_0 = y$$

$$x(y - \hat{y}) = 0$$

$$x^T(y - \hat{y}) = 0$$

$$\hat{y} = x \hat{\theta}$$

$$x^T(y - x \hat{\theta}) = 0$$

x_0 is always 1 cuz it is the term of intercept and it is always constant.

$$h(x) = \sum \theta_i x_i = \theta^T x$$